Indefinite integration

Basic Rules of integrations:

$$\int a\,x^n\,dx = \frac{a}{n+1}\,x^{n+1} + c$$

1) $\int (x^3 + 3x^2 + 5)dx = \frac{x^4}{4} + \frac{3}{3}x^3 + 5x + c$

2) $\int \left(x^{\frac{1}{3}} + x^{-3}\right)dx = \frac{1}{\frac{4}{3}}x^{\frac{4}{3}} + \frac{1}{-2}x^{-2} + c = \frac{3}{4}x^{\frac{4}{3}} - \frac{1}{2}x^{-2} + c$

$$\int [f(x)]^n\,f'(x)\,dx = \frac{1}{n+1}\,[f(x)]^{n+1} + c$$

1) $\int 2x(x^2+3)^5 dx = \frac{(x^2+3)^6}{6} + c = \frac{1}{6}(x^2+3)^6 + c$

2) $\int x^3(x^4-5)^3 dx = \frac{1}{4}\int 4x^3(x^4-5)^3 dx = \frac{1}{4}\frac{1}{4}\,(x^4-5)^4 + c = \frac{1}{16}(x^4-5)^4 + c$

3) $\int \cos x \sin^6 x\ dx = \frac{1}{7}\sin^7 x + c$

$$\int \frac{f'(x)}{f(x)}\,dx = \ln[f(x)] + c$$

1) $\int \frac{x+1}{x^2+2x}\,dx = \frac{1}{2}\int \frac{2x+2}{x^2+2x}\,dx = \frac{1}{2}\ln(x^2+2x) + c$

2) $\int \frac{\sec^2 x}{3+\tan x}dx = \ln(3+\tan x) + c$

3) $\int \frac{2-3x^2}{x^3-2x}dx = -\int \frac{3x^2-2}{x^3-2x}dx = -\ln(x^3-2x) + c$

$$\int \frac{f'(x)}{\sqrt{f(x)}}\,dx = 2\sqrt{f(x)} + c$$

1) $\int \frac{x^3-1}{\sqrt{x^4-4x}}dx = \frac{1}{4}\int \frac{4x^3-4}{\sqrt{x^4-4x}}dx = \frac{1}{4}\ 2\sqrt{x^4-4x}+c = \frac{1}{2}\sqrt{x^4-4x}+c$

$$\int f'(x)\cos[f(x)]\ dx = \sin[f(x)]+c$$

1. $\int \sin x\ dx = -\cos x + c$	2. $\int \cos x\ dx = \sin x + c$
3. $\int \sec^2 x\ dx = \tan x + c$	4. $\int \operatorname{cosec}^2 x\ dx = -\cot x + c$
5. $\int \sec x \tan x\ dx = \sec x + c$	6. $\int \operatorname{cosec} x \cot x\ dx = -cosec\ x + c$

1) $\int x\cos x^2\ dx = \frac{1}{2}\int 2x\cos x^2\ dx = \frac{1}{2}\sin x^2 + c$

2) $\int \tan x\ dx = -\int \frac{-\sin x}{\cos x}dx = -\ln(\cos x)+c = \ln\sec x + c$

3) $\int \sec^2 x\ dx = \tan x + c$

4) $\int(\sin 4x + \cos 6x)dx = -\frac{1}{4}\cos 4x + \frac{1}{6}\sin 6x + c$

$$\int f'(x)\ e^{f(x)}dx = e^{f(x)}+c$$

1) $\int(e^{3x}+e^{2x})dx = \frac{1}{3}e^{3x}+\frac{1}{2}e^{2x}+c$

2) $\int x\ e^{2x^2}dx = \frac{1}{4}\int 4x\ e^{2x^2}dx = e^{2x^2}+c$

$$\int \frac{u'}{a^2+u^2}dx = \frac{1}{a}\tan^{-1}\frac{u}{a}+c$$

$$\int \frac{u'}{\sqrt{a^2-u^2}}dx = \sin^{-1}\frac{u}{a}+c$$

1) $\int \frac{1}{25+x^2}\,dx = \int \frac{1}{(5)^2+(x)^2}\,dx = \frac{1}{5}\tan^{-1}\frac{x}{5} + c$

2) $\int \frac{1}{16+9x^2}\,dx = \frac{1}{3}\int \frac{3}{(4)^2+(3x)^2}\,dx = \frac{1}{3}\frac{1}{4}\tan^{-1}\frac{3x}{4} + c = \frac{1}{12}\tan^{-1}\frac{3x}{4} + c$

3) $\int \frac{1}{\sqrt{16-x^2}}\,dx = \int \frac{1}{\sqrt{(4)^2-(x)^2}}\,dx = \sin^{-1}\frac{x}{4} + c$

4) $\int \frac{1}{\sqrt{16-25x^2}}\,dx = \frac{1}{5}\int \frac{5}{\sqrt{(4)^2-(5x)^2}}\,dx = \frac{1}{5}\sin^{-1}\frac{5x}{4} + c$

Integration by parts

$$\int u\,dv = u\,v - \int v\,du$$

Examples

1) $\int x \sin x\ dx$

$u = x$	$dv = \sin x\ dx$
$du = dx$	$v = -\cos x$

$$I = -x\cos x + \int \cos x\ dx = -x\cos x + \sin x + c$$

2) $\int x \sin x\ dx$

u		dv
x	↘ +	$\sin x$
1	↘ −	$-\cos x$
0		$-\sin x$

$$I = -x\cos x + \sin x + c$$

3) $\int x\,e^x\,dx$

Soln

$$I = \int x\,e^x\,dx$$

$$u = x \qquad dv = e^x dx$$

$$du = dx \qquad v = e^x$$

$$I = u\,v - \int v\,du = x\,e^x - \int e^x\,dx$$

$$I = x\,e^x - e^x + c$$

4) $\int x^3 \cos x \ dx$

Soln

$$I = \int x^3 \cos x \, dx$$

$u = x^3$	↘ +	$dv = \cos x \ dx$
$3x^2$	↘ −	$\sin x$
$6x$	↘ +	$-\cos x$
6	↘ −	$-\sin x$
0		$\cos x$

$$I = x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + c$$

5) $\int \ln x \ dx$

$u = \ln x$	$dv = \ dx$
$du = \frac{1}{x} dx$	$v = x$

$$I = x \ln x - \int x \frac{1}{x} \, dx = x \ln x - \int dx = x \ln x - x + c$$

6) $\int x^2 \ln x \ dx$

$u = \ln x$	$dv = x^2 \ dx$
$du = \frac{1}{x} dx$	$v = \frac{x^3}{3}$

$$I = \frac{1}{3} x^3 \ln x - \int \frac{x^3}{3} \frac{1}{x} \, dx = \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 \, dx = \frac{1}{3} x^3 \ln x - \frac{1}{3} \frac{1}{3} x^3 + c$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + c$$

Integration of rational functions (Partial Fractions)

1) $\int \frac{x}{x^2+2x}\,dx$

$$\int \frac{x}{x^2+2x}\,dx = A\int \frac{1}{x}dx + B\int \frac{1}{x+2}\,dx = A\ln x + B\ln(x+2) + c =$$
$$= \frac{1}{2}\ln x + \frac{1}{2}\ln(x+2) + c$$

To find the values of constants

$$\frac{x+1}{x^2+2x} = \frac{x+1}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2} = \frac{A(x+2)+Bx}{x(x+2)}$$

$$A(x+2) + Bx = x + 1$$

Put $x = 0$ $\quad A(2) + B(0) = 0 + 1 \quad 2A = 1 \quad A = \frac{1}{2}$

Put $x = -2$ $\quad A(0) + B(-2) = -2 + 1$

$$-2B = -1 \qquad B = \frac{1}{2}$$

$$\int \frac{x}{x^2+2x}\,dx = A\int \frac{1}{x}dx + B\int \frac{1}{x+2}\,dx = A\ln x + B\ln(x+2) + c =$$
$$= \frac{1}{2}\ln x + \frac{1}{2}\ln(x+2) + c$$

2) $\int \frac{17-4x}{(x-2)(x+1)}\,dx$

$$\frac{17-4x}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$$

$$\int \frac{17-4x}{(x-2)(x+1)}dx = A\int \frac{1}{x-2}dx + B\int \frac{1}{x+1}dx = A\ln(x-2) + B\ln(x+1) + c$$
$$= 3\ln(x-2) - 7\ln(x+1) + c$$

To find the values of constants

$$A(x+1) + B(x-2) = 17 - 4x$$

Put $x = 2$ $\quad A(3) = 17 - 8 \quad 3A = 9 \quad A = 3$

Put $x = -1 \quad B(-3) = 17 + 4 \quad -3B = 21 \quad B = -7$

3) $\int \frac{5x-2}{(x+3)^2}dx$

$$\frac{5x-2}{(x+3)^2} = \frac{A}{x+3} + \frac{B}{(x+3)^2}$$

$$\int \frac{5x-2}{(x+3)^2}dx = A\int \frac{1}{x+3}dx + B\int \frac{1}{(x+3)^2}dx = A\int \frac{1}{x+3}dx + B\int (x+3)^{-2}dx$$
$$= A\ln(x+3) - B\,(x+3)^{-1} + c$$

$$\frac{5x-2}{(x+3)^2} = \frac{A}{x+3} + \frac{B}{(x+3)^2}$$

$$A(x+3) + B = 5x - 2$$

Put $x = -3 \quad B = -15 - 2 = -17$

Put $x = 0 \quad 3A + B = -2 \quad 3A - 17 = -2$

$$3A = 15 \quad A = 5$$

4) $\int \frac{4}{x(x+2)^2}dx$

$$\frac{4}{x(x+2)^2} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

$$\int \frac{4}{x(x+2)^2}\,dx = A\int \frac{1}{x}dx + B\int \frac{1}{x+2}dx + C\int \frac{1}{(x+2)^2}dx$$
$$= A\ln x + B\ln(x+2) - C(x+2)^{-1} + c$$

$$\frac{4}{x(x+2)^2} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

$$A(x+2)^2 + Bx(x+2) + Cx = 4$$

Put $x = 0$ $\quad 4A = 4 \quad A = 1$

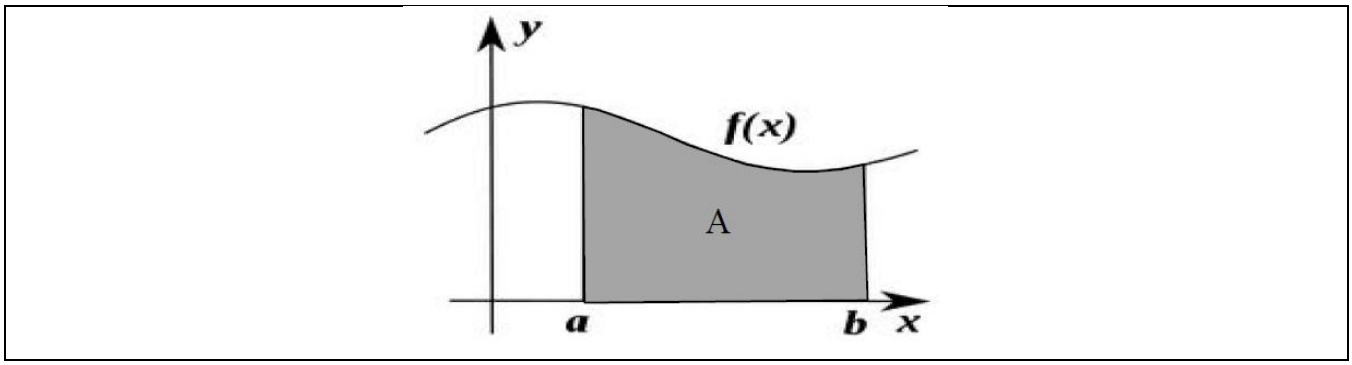
Put $x = -2 \quad -2C = 4 \quad C = -2$

Put $x = 1 \quad A(9) + B(3) + C = 4$

$$9 + 3B - 2 = 4 \qquad 3B = -3 \qquad B = -1$$

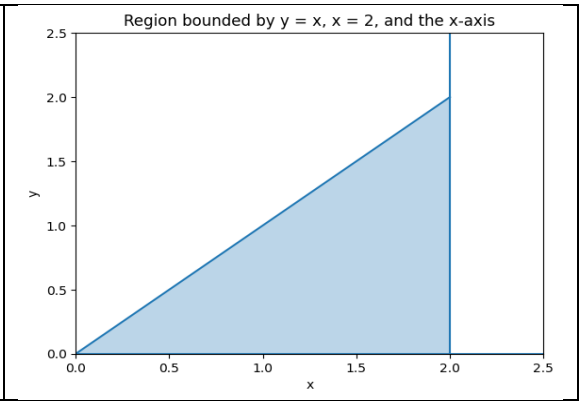
Definite integrations:

$$\int_1^2 3x^2\,dx = [x^3]_1^2 = (2^3) - (1^3) = 8 - 1 = 7$$



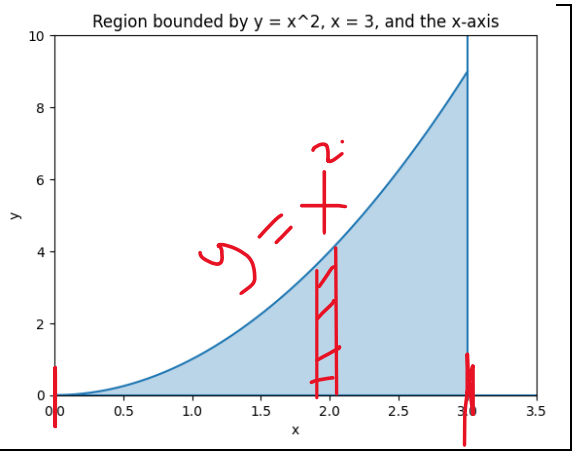
$$\int_a^b [f(x) - 0]dx$$

Find the area bounded by the lines $y = x,\ x = 2 \quad and\ x - axis$



$$\int_0^2 [x - 0]dx = \left[\frac{x^2}{2}\right]_0^2 = \left(\frac{2^2}{2}\right) - (0) = 2\ square\ unit$$

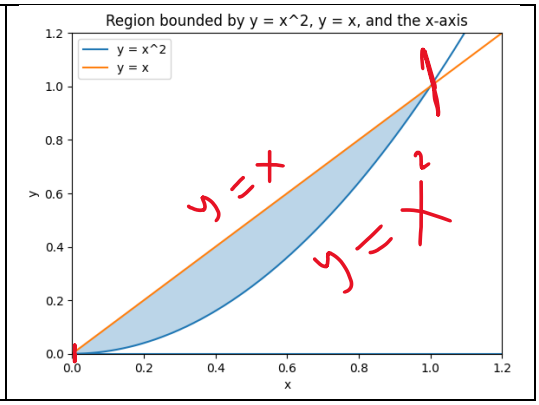
Find the area bounded by the lines $y = x^2, x = 3 \quad and\ x - axis$



$y = 0$

$$\int_0^3 [x^2 - 0]dx = \left[\frac{x^3}{3}\right]_0^3 = \left(\frac{3^3}{3}\right) - (0) = 9\ square\ unit$$

Find the area bounded by the lines $y = x^2,\ y = x \quad and\ x - axis$

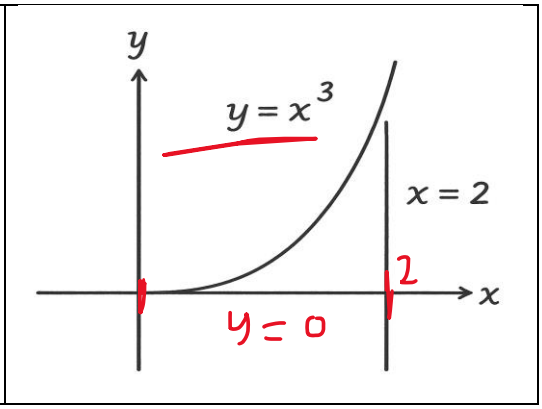


$$x = x^2 \quad x^2 - x = 0 \quad x(x-1) = 0$$

$$x = 0 \qquad x = 1$$

$$\int_0^1 [x - x^2]dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \left(\frac{1^2}{2} - \frac{1^3}{3}\right) - (0) = \frac{1}{6}\ square\ unit$$

Find the area bounded by the lines $y = x^3,\ x = 2 \quad and\ x - axis$



$$\int_0^2 [x^3 - 0]dx = \left[\frac{x^4}{4}\right]_0^2 = \left(\frac{2^4}{4}\right) - (0) = 4\ square\ unit$$

Linear transformations

If V and W are vector spaces and F is a function that associates a unique vector in W with each vector in V, we say F **maps** V into W, and write $F: V \to W$. Further if F associates the vector W with the vector v, we write $\mathrm{w} = F(\mathrm{v})$ and say that w is the **image** of v under F.

vector form R^2 $\quad u = (x\,, y)$

Matrix form $u = \begin{bmatrix} x \\ y \end{bmatrix}$

$$R^3 \qquad u = (x\,, y\,, z) = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$T: R^n \to R^m$$

$$T(x\,, y\,, z) = (x + 2y\,, y - 2z)$$

$$T: v \to w$$

v preimage

w image

Example 1:

If $T: R^2 \to R^2$, then $\mathrm{v} = (x\,, y) \in R^2$ and $T(x\,, y) = (x - y\,, x + 2y)$

(a) find the image of $\mathrm{v} = (-1\,, 2)$

(b) find the preimage of $\mathrm{w} = (-1\,, 11)$

$$T(x\,, y) = (x - y\,, x + 2y)$$

$$T(-1\,, 2) = (-1 - 2\,, -1 + 4) = (-3\,, 3)$$

$$T(x\,,y) = (x - y\,, x + 2y) = (-1\,, 11)$$

$$x - y = -1 \qquad x = y - 1$$

$$x + 2y = 11 \qquad y - 1 + 2y = 11 \quad 3y = 12 \quad y = 4$$

$$x = y - 1 \qquad x = 3$$

Preimage is (3 , 4)

Linear transformation

I. $T(u + \mathrm{v}) = T(u) + T(\mathrm{v})$ for all vectors u and v in V.

II. $T(ku) = kT(u)$ for all vectors u in V and all scalars k.

Example:

Let T: $R^2 \to R^2$ be the function defined by $T(x\,,y) = (x - y\,, x + 2y)$. Show that this transformation is linear or not

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - y \\ x + 2y \end{bmatrix}$$

$$T(x\,,y) = (x - y\,, x + 2y)$$

Put $u = (x_1\,, y_1)$ and $v = (x_2\,, y_2)$

$$u + v = (x_1 + x_2\,, y_1 + y_2)$$

$$T(u + v) = T(x_1 + x_2\,, y_1 + y_2)$$

$$= \big(x_1 + x_2 - (y_1 + y_2)\,, x_1 + x_2 + 2(y_1 + y_2)\big)$$

$$= (x_1 + x_2 - y_1 - y_2\,, x_1 + x_2 + 2y_1 + 2y_2)$$

$$= (x_1 - y_1 + x_2 - y_2\,, x_1 + 2y_1 + x_2 + 2y_2)$$

$$= (x_1 - y_1\,, x_1 + 2y_1\,) + (x_2 - y_2\,, x_2 + 2y_2)$$

$$= T(x_1, y_1) + T(x_2, y_2)$$

$$= T(u) + T(v)$$

$$ku = k(x_1, y_1) = (kx_1, ky_1)$$

$$T(kx_1, ky_1) = (kx_1 - ky_1, kx_1 + 2ky_1) = k(x_1 - y_1, x_1 + 2y_1) = kT(x_1, y_1)$$

The transformation is linear

A linear transformation defined by a matrix

$$T(\mathrm{v}) = A\mathrm{v}$$

A standard Matrix

$$R^2 \qquad e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$R^3 \qquad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Example: Determine the standard matrix of $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ 3y \end{bmatrix}$

$$T(e_1) = T\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$T(e_2) = T\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

Ex2: Find the standard matrix for the linear transformation T: $R^3 \rightarrow R^2$ defined by

$$T(x\,,y\,,z) = (x - 2y\,,2x + y)$$

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x - 2y \\ 2x + y \end{bmatrix}$$

$$T\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad T\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \qquad T\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{bmatrix}$$

Kernel

The set of vectors in U that are mapped into the zero vector of V is called the kernel (or nullspace) of T.

Example:

Let T: $R^3 \rightarrow R^2$ be a linear transformation defined by $T(x\,,y\,,z) = (x + 2y - z\,,-2x - 4y + 2z)$. Which is the following are in $\ker(T)$?

a) $(3\,,-2\,,-1)$

b) $(1\,,1\,,2)$

$$T(3\,,-2\,,-1) = (3 - 4 + 1\,,-6 + 8 - 2) = (0\,,0)$$

$$T(1\,,1\,,2) = (1 + 2 - 2\,,-2 - 4 + 4) = (1\,,-2)$$

Example:

Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ find the kernel of linear transformation $T: R^2 \rightarrow R^2$?

solution

$$T(v) = Av = 0$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x + 3y \\ 2x + 6y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x + 3y = 0$$

$$2x + 6y = 0$$

Put $y = t \qquad x = -3t$

The kernel is $\begin{bmatrix} -3t \\ t \end{bmatrix} = t \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

Dimension (n)

Rank of T = rank of Matrix

Nullity

$n = (rank\ of\ T) + (nullity\ of\ T)$

$$nullity = n - rank$$

Example:

Find the rank and nullity of the linear transformation $T: R^3 \longrightarrow R^3$ define by

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$n = 3$

Rank $= 2$

Nullity$= 3 - 2 = 1$

Example:

Find the rank and nullity of the linear transformation $T: R^2 \longrightarrow R^2$ define by

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$n = 2$

Rank $= 2$

Nullity$= 2 - 2 = 0$

Example:

Find the rank and nullity of the linear transformation $T: R^3 \longrightarrow R^3$ define by

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Solution

$$rank\ T = rank\ A = 2$$

$$n = 3$$

$$nullity\ of\ T = n - rank\ of\ T = 3 - 2 = 1$$

Example:

Consider the basis $S = \{u_1 , u_2 , u_3\}$ for R^3, where

$u_1 = (1 , 1 , 1), u_2 = (1 , 1 , 0) , u_3 = (1 , 0 , 0)$, and let $T: R^3 \rightarrow R^2$ be a linear transformation such that

$$T(u_1) = (1 , 0) \qquad T(u_2) = (2 , -1) \quad T(u_3) = (4 , 3)$$

Find $T(2 , -3 , 5)$

$$(2 , -3 , 5) = K_1 u_1 + K_2 u_2 + K_3 u_3$$

$$(2 , -3 , 5) = K_1(1 , 1 , 1) + K_2(1 , 1 , 0) + K_3(1 , 0 , 0)$$

$$(2 , -3 , 5) = (K_1 , K_1 , K_1) + (K_2 , K_2 , 0) + (K_3 , 0 , 0)$$

$$K_1 + K_2 + K_3 = 2$$

$$K_1 + K_2 + 0 = -3$$

$$K_1 + 0 + 0 = 5$$

$$K_1 = 5$$

$$5 + K_2 = -3 \quad K_2 = -8$$

$$5 - 8 + K_3 = 2 \qquad K_3 = 5$$

$$(2 , -3 , 5) = 5(1 , 1 , 1) - 8(1 , 1 , 0) + 5(1 , 0 , 0)$$

$$T(2 , -3 , 5) = 5T(1 , 1 , 1) - 8T(1 , 1 , 0) + 5T(1 , 0 , 0)$$

$$T(2 , -3 , 5) = 5(1 , 0) - 8(2 , -1) + 5(4 , 3) = (5 , 0) + (-16 , 8) + (20 , 15)$$
$$= (5 - 16 + 20 , 0 + 8 + 15)$$
$$= (9 , 23)$$

- Ex:

The L.T. $T: R^n \to R^m$ is given by $T(\mathbf{x}) = A\mathbf{x}$, find the nullity and rank of T and determine whether T is one - to - one, onto, or neither.

$$(a)\ A=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix} \qquad (b)\ A=\begin{bmatrix}1 & 2\\ 0 & 1\\ 0 & 0\end{bmatrix}$$

$$(c)\ A=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & -1\end{bmatrix} \qquad (d)\ A=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 0 & 0 & 0\end{bmatrix}$$

Sol:

$T: R^n \to R^m$	dim(domain of T)	rank(T)	nullity(T)	1-1	onto
(a) $T: R^3 \to R^3$	3	**3**	**0**	**Yes**	**Yes**
(b) $T: R^2 \to R^3$	2	2	**0**	**Yes**	No
(c) $T: R^3 \to R^2$	3	**2**	1	No	**Yes**
(d) $T: R^3 \to R^3$	3	2	1	No	No

Relations and their properties

Properties of Relations

Reflexive

لو كل عنصر داخل المجموعة ارتبط مع نفسه

$$s = \{1 , 2 , 3\}$$

$$R = \{(1 , 1) , (2 , 2) , (3 , 3) , (1 , 3)\}$$

Symmetric

اذا تواجدت العلاقة ومقلوبها

if $(b, a) \in R$ whenever $(a, b) \in R$

$R_2 = \{(1 , 1), (1 , 2), (2 , 1)\}$

$$R_3 = \{(1 , 1), (1 , 2), (1 , 4), (2 , 1), (2 , 2), (3 , 3), (4 , 1), (4 , 4)\}$$

Transitive

اذا كانت العلاقات مغلقة بين جميع العناصر

$R_4 = \{(2 , 1), (3 , 1), (3 , 2), (4 , 1), (4 , 2), (4 , 3)\}$

$R_5 = \{(1 , 1), (1 , 2), (1 , 3), (1 , 4), (2 , 2), (2 , 3), (2, 4), (3 , 3), (3 , 4), (4 , 4)\}$

$R_6 = \{(3 , 4)\}$

Equivalence Relation

A relation is an **equivalence relation** if it is **reflexive, symmetric, and transitive**.

Combining relations

Union of Relations

Let $R = \{(1,2), (2,3)\}$ and $S = \{(2,3), (3,4)\}$. Then,

$$R \cup S = \{(1,2), (2,3), (3,4)\}$$

Intersection of Relations

Let $R = \{(1,2), (2,3)\}$ and $S = \{(2,3), (3,4)\}$. Then,

$$R \cap S = \{(2,3)\}$$

Difference of Relations

Let $R = \{(1,2), (2,3)\}$ and $S = \{(2,3), (3,4)\}$. Then,

$$R - S = \{(1\ , 2)\}$$

$$S - R = \{(3\ , 4)\}$$

Composition of Relations

Let $R = \{(1,2), (2,3)\}$ and $S = \{(2,3), (3,4)\}$. Then,

$$R \circ S = \{(1\ , 3)\ , (2\ , 4)\ \}$$

Power of a Relation

$$R^2 = R \circ R$$

$$R^3 = R^2 \circ R$$

Let $R = \{(1,2), (2,3)\}$ $R = \{(1,2), (2,3)\}$

$$R^2 = R \circ R = \{(1\ , 3)\}$$

Complement of a Relation

If $S = \{1\ ,2\ ,3\}$ and $R = \{(1\ ,2), (2\ ,3)\}$, then

$$R^R = \{(1\ ,2)\ ,(2\ ,3)\ ,(1\ ,1)\ ,(2\ ,2)\ ,(3\ ,3)\} = R \cup \{\ (1\ ,1)\ ,(2\ ,2)\ ,(3\ ,3)\}$$

$$R^S = \{(1\ ,2)\ ,(2\ ,3)\ ,(2\ ,1)\ ,(3\ ,2)\} = R \cup \{\ (2\ ,1)\ ,(3\ ,2)\}$$

$$R^T = \{(1\ ,2)\ ,(2\ ,3)\ ,(1\ ,3)\} = R \cup \{\ (1\ ,3)\}$$

Difference symmetric

$$R \oplus S = (R \cup S) - (R \cap S)$$

Let $R = \{(1,2), (2,3)\}$ and $S = \{(3,4), (2,3)\}$.

Then $R \oplus S =$

$$R \cup S = \{(1\ ,2)\ ,(2\ ,3)\ ,(3,4)\}$$

$$R \cap S = \{(2\ ,3)\}$$

$$R \oplus S = \{(1\ ,2)\ ,(3\ ,4)\}$$

Let $u = \{1\ ,2\ ,3\}$ $R = \{(1,2), (2,3)\}$ and $S = \{(2,3), (3,2)\}$. Then,

$$R \circ S = \{(1\ ,3)\ ,(2\ ,2)\ \}$$

$$R \circ S = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1 2 3

$$R \cup S = \{(1, 2), (2\ ,3)\ ,(3\ ,2)\}$$

1

3

$$R \cup S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$